

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• C	1
01.2	• Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms • Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas	2
01.3	• HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere • So most do not persist long enough to reach the stratosphere where UV would release their chlorine	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
02.1	• Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier • Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground • Take repeated readings at each distance and calculate means (and standard deviation) to assess variability • Use a measure such as LA10 or Leq over a set period rather than a single peak reading • Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance	5
02.2	• Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier	1

Question	Marking guidance	Marks
02.3	• LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods • LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level	2
02.4	<i>Any 2 from:</i> • Double / triple glazing • Acoustic insulation in walls / roof • Bedrooms on the side facing away from the track • Dense hedges / trees and earth mounds in gardens	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
03.1	• Readings drift so they may not be accurate; calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions • pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly • If a meter cannot be adjusted, record the percentage error and correct the results	3
03.2	• Same height above the ground • No obstacles to air flow; impeller type facing into the wind • Axis of rotation horizontal (impeller) or vertical (cup); same time period / averaging	3
03.3	• Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling • Evaporation cools the wet bulb more when humidity is low; the temperature difference is compared with a table to find humidity	2
03.4	<i>Any 2 from:</i> • Test strips with reagent bands compared with a colour chart (semi-quantitative) • Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength • Ion-selective electrode measuring electrode potential (other ions can interfere)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
04.1	• Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes • Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH • Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow • Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser • Unattractive topography - landscaping / reshaping • Dust / heavy metals - cover and keep at high pH	6
04.2	• So that plant roots cannot reach the toxic layer • Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains	2

Question	Marking guidance	Marks
04.3	- Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions - Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
05.1	$(210 - 12) / 210 \times 100$ = 94.3% (accept 94%)	2
05.2	<i>Any 3 from:</i> - Conserves mineral reserves / extends their lifetime - Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water) - Reduces mineral processing pollution (eg red mud from bauxite refining) - Reduces waste disposal impacts / landfill volume - Less CO₂ / climate change	3
05.3	<i>Any 4 from:</i> - Identification of materials: slow, labour-intensive to determine composition where it matters (eg alloys, plastic types) - Separation of mixed materials: alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed; mixed wastes reduce quality - Reduction in quality / contamination (eg paper mixed with food waste) - Transport / collection: small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings) - Lack of consumer cooperation in separating wastes; waste losses to litter and general waste; labour costs	4
05.4	Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7.	
06.1	- Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared - Greenhouse gases (naturally water vapour and CO₂) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface - This keeps the surface about 33 °C warmer than it would otherwise be	3
06.2	- CO₂ is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest - CO₂ has a long residence time so it accumulates; methane has a short residence time (12 years) so it is removed faster - CFC emissions are now very small following the Montreal Protocol	3
06.3	- Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect - Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut	2

Question	Marking guidance	Marks
06.4	<i>Any 2 from:</i> • Collect landfill / anaerobic digester gas and use as fuel • Changes in cattle diet / fewer cattle / less meat consumption • Draining rice paddies periodically • Anaerobic digestion of manure with gas capture	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	
07.1	• Cost increases as the percentage prevented increases • The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90%	2
07.2	• $16.0 - 5.0 = 11.0$ (£ million)	1
07.3	• ALARA: emissions should be As Low As Reasonably Achievable • It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set • BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive • The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99%	4
07.4	<i>Any 3 from:</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen • Better understanding of the health / environmental effects so stricter standards are set • Improved technology becomes available (BATNEEC changes) / cost of better control falls • Pollutant may have accumulated (persistent) so background levels are higher	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
08.1	<i>Any 3 from:</i> • Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) • Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles • Carbon neutral in principle: CO₂ released equals CO₂ absorbed in photosynthesis • Energy density of alcohol / biodiesel nearly as high as fossil fuels; wastes (straw, sewage, landfill gas) are cheap	3
08.2	<i>Any 3 from:</i> • Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) • Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO₂ as using fossil fuels directly • Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally • Supply from wastes limited to the amount produced	3

Question	Marking guidance	Marks
08.3	<i>Any 2 from:</i> - Hydrogen from algae: some algae deprived of sulfur produce hydrogen during photosynthesis - Anaerobic digestion extended to crop wastes and marine algae; small-scale digesters (eg one cow's manure supplying a family's cooking gas) - Biofuels from microorganisms (eg oil-producing microalgae)	2
08.4	- Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK - New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	
09.1	- Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) - They work best in sheltered / calm water such as the lagoon - Limitation: they do not work in open water where currents or waves are strong; oil can flow under the skirt in a strong current	3
09.2	- Skimmers: rotating (metal) discs / belts pick up floating oil which is scraped off and stored - Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed - Collected oil / absorbents are incinerated or sent to landfill (or oil recycled)	3
09.3	- Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface - They do not reduce the amount of oil; the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water	2
09.4	- Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) - Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9
10.2	• The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Question	Marking guidance	Marks
11.1	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25